

Unit-2 Research Topic, Literature Survey and Problem Statement

Basanta Chapagain, Patan Multiple Campus

Research Theme

- A **research theme** is the broad area or overarching direction of a research study. It represents a major subject or field of knowledge within which multiple specific research topics can be explored and developed.
- A research theme usually reflects current trends, emerging interests, or significant challenges in a particular discipline. It serves as a general framework that allows researchers to investigate various issues connected to the same core idea.

Examples of Research Themes : Digital Transformation in Education , Human–Computer Interaction, Artificial Intelligence and Machine Learning, Cybersecurity and Data Privacy, Sustainable Information Technology

Research Theme

Characteristics of a Research Theme

- Very broad and general in scope
- Encompasses many related topics and sub-areas
- Commonly used in academic conferences, journals, and research funding programs
- Helps organize and group similar studies under one unified concept

Research Theme

Example in the Field of Computer Applications

- **Theme:** Cybersecurity

Possible Research Topics under this theme:

- Network security in cloud computing environments
- Development of phishing attack detection systems using machine learning
- Security vulnerabilities and protection mechanisms in mobile banking applications

Research Topic

- A **research topic** is the **specific subject or area that a researcher wants to study**. It is narrower than the theme and identifies **what exactly the researcher wants to investigate**.
- The topic provides the **direction and scope** of the research work.
- Choosing a research topic is often the most underrated hurdle in the academic process. It's the foundation of your entire project

“If the foundation is weak, the entire structure cannot remain stable—no matter how well the architecture or methodology is designed.”

Research Topic

Example Topics (Computer Applications)

- Impact of cloud computing on business data storage
- Use of mobile applications for online learning
- Performance analysis of web-based database systems
- Role of artificial intelligence in healthcare systems

Research Topic

Characteristics of a Good Research Topic

A research topic should be:

Clear and Focused

- In tech, "broad" topics lead to shallow results. You must define the specific technology, the user base, and the intended outcome.
- **The "Too Broad" Version:** "Artificial Intelligence in Healthcare."
- **The "Good" Topic:** "Comparing the accuracy of **Convolutional Neural Networks (CNN)** vs. **Vision Transformers** in detecting early-stage melanoma from smartphone images."
- **Why it works:** It identifies the specific algorithms, the exact medical condition, and the hardware constraints.

Research Topic

Interesting to the Researcher

- The field of Computer Applications moves fast. You should choose a sub-sector (e.g., Cybersecurity, UX/UI, Cloud Computing) that aligns with your career goals.
- **The Motivation:** If you are passionate about gaming, researching "**Latency reduction algorithms in Cloud Gaming**" will be much more rewarding than researching database management systems.
- **The Persistence Factor:** When your code refuses to compile or your model won't converge, your interest in the subject is what keeps you debugging.

Research Topic

Relevant to the Field of Study

- A good topic addresses "The State of the Art." It should tackle current bottlenecks or emerging paradigms in computing.
- **The "Knowledge Gap":** Many apps exist for productivity, but perhaps there is a gap in **"Cross-platform synchronization for AR-based collaborative workspaces."**
- **Industry Alignment:** Researching **"Edge Computing for IoT in Smart Agriculture"** is highly relevant as industry shifts away from centralized cloud processing to save bandwidth.

Research Topic

Feasible (Resources and Time)

- Feasibility in computing often comes down to hardware, data, and API access.
- **Data Availability:** You cannot research "Predicting stock market crashes with AI" if you don't have access to high-frequency trading datasets or the GPU power to process them.
- **Technical Scope:** Developing a full-scale decentralized social media platform might be too much for a 3-month project. However, developing a "**Proof-of-concept Smart Contract for secure user authentication on the Ethereum blockchain**" is feasible.

Research Topic

Supported by Existing Literature

- Even if you are using a brand-new framework (like the latest version of Flutter or a new LLM), your research must be grounded in established computer science principles.
- **Theoretical Grounding:** Your work on a new mobile app should be supported by existing literature on **Human-Computer Interaction (HCI)** or **Agile Development Methodologies**.
- **Benchmarking:** To prove your application is "good," you need existing studies to provide benchmarks (e.g., "The industry standard for app load time is \$2/ seconds; your proposed optimization reduced it to \$1.2/ seconds").

Research Issue / Research Problem

- A **research issue or research problem** refers to the **specific difficulty, gap, or unanswered question** within the research topic that needs investigation.
- It identifies **what problem exists and why the research is necessary**.
- The **research problem is the core focus of the study** because the research is conducted to **find solutions or explanations for this problem**.

A research problem usually arises when:

- Existing studies show **limitations**
- There is **conflicting evidence**
- A practical problem requires **scientific investigation**

Research Issue / Research Problem

Characteristics of a Good Research Problem

- Clearly defined
- Researchable and measurable
- Relevant to the field
- Significant enough to contribute knowledge
- Feasible to investigate

Research Issue / Research Problem :Example

Topic: **Online learning systems in universities**

Possible research problems:

- Low student engagement in virtual classrooms
- Difficulty in assessing student performance online
- Lack of interaction between teachers and students in e-learning

• Topic: **Mobile Banking Applications**

Possible research problems:

- Security risks in mobile banking systems
- Lack of user trust in digital payment systems

Research Issue / Research Problem :Example

Example in Computer Applications

- Theme: **Human–Computer Interaction**
- Topic: **User interface design for mobile applications**
- Research Problem: **How poor interface design affects user experience in mobile learning apps**

Research Issue / Research Problem :Example

Importance of Identifying Research Topic/Issue/Theme

- Identifying the research theme, topic, and issue is important because it:
- Provides **clear direction for the study**
- Helps define the **scope and objectives**
- Guides the **selection of research methods**
- Helps in **reviewing relevant literature**
- Ensures the research **addresses real problems**
- Without clearly identifying these elements, research may become **unfocused and ineffective.**

Research Issue / Research Problem :Steps

Step 1: Choose a Broad Area of Interest

- First, select a **general field** that interests you or is related to your academic discipline.

Examples in Computer Applications

- Artificial Intelligence
- Web Development
- Cybersecurity
- E-learning Systems
- Mobile Application Development

Example

- Broad Area: **E-Learning Systems**

Research Issue / Research Problem :Steps

Step 2: Conduct Preliminary Reading (Literature Review)

- Read **journals, conference papers, articles, and books** to understand:
- What research already exists
- Current trends and technologies
- Problems researchers have identified
- This helps you **avoid repeating existing research** and find areas that need further study.

Example

- After reading articles on e-learning systems, you may find that:
- Many universities use **Learning Management Systems (LMS)**.
- Students sometimes show **low engagement in online classes**.

Research Issue / Research Problem :Steps

Step 3: Identify Gaps or Issues

- A **research gap** is something that existing research has not fully addressed.
- Look for:
 - Limitations in previous studies
 - Unanswered questions
 - New technologies not yet studied
 - Problems occurring in real practice

Example

- Gap identified:
- Many studies discuss e-learning adoption, but **few examine student engagement in programming courses delivered online.**

Research Issue / Research Problem :Steps

Step 4: Narrow Down the Topic

- Convert the general area into a **more focused topic**.

Example

- Broad area: E-learning systems
Focused topic: **Online learning platforms for programming courses**

Research Issue / Research Problem :Steps

Step 5: Define the Specific Research Problem

- Now convert the topic into a **clear research problem or issue**.

Ask questions such as:

- What exactly is the problem?
- Why does it occur?
- Who is affected?

Example Research Problem

- Students learning programming through online platforms often show **low engagement and difficulty understanding practical coding concepts**.

Research Issue / Research Problem :Steps

Step 6: Formulate Research Questions

- Transform the research problem into **specific research questions**.

Example Research Questions

- How does the design of an online learning platform affect student engagement in programming courses?
- What features improve student participation in online coding classes?

Research Issue / Research Problem :Steps

Step 7: Check Feasibility

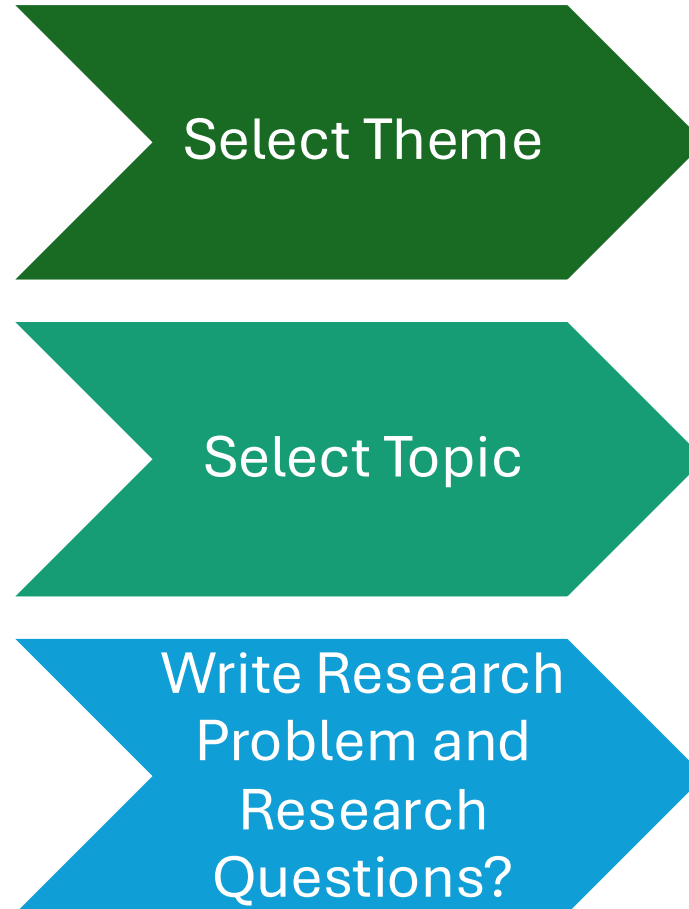
Ensure the problem is **researchable** with available:

- Time
- Data
- Resources
- Technical tools

Example

- You can collect data by:
- Surveying computer science students
- Analyzing usage data from an LMS
- Conducting interviews with instructors

Research Issue / Research Problem : Tasks



Research Issue / Research Problem :Criteria

- **Researcher's Interest:** The research problem should be interesting to the researcher so that they remain motivated to study it.
- **Relevance to the Field:** The problem should be related to the researcher's academic or professional field of study.
- **Availability of Data:** Adequate data and information should be available to investigate the research problem.
- **Feasibility:** The research problem should be manageable within the available time, resources, and skills.
- **Researchability:** The problem must be capable of being studied using scientific research methods.

Research Issue / Research Problem :Criteria

- **Significance:** The research problem should contribute useful knowledge or help solve a real-world issue.
- **Clarity and Specificity:** The research problem should be clearly defined and not too broad or vague.
- **Originality:** The research problem should provide new insights or add value to existing knowledge.
- **Ethical Acceptability:** The research should follow ethical principles and not harm participants.
- **Social or Practical Value:** The research problem should have benefits for society, organizations, or the academic community.

Literature Review

- The most important preliminary task to undertake a research study is to pursue the existing literature available in the field of interest.
- The survey of literature and review of literature is an integral part of the research to acquaint with the existing knowledge or material available in the area of interest.
- Literature survey means penetrating through the available literature in the selected area of research, whereas, literature review involves assessment and examination of the reported contributions/findings.

Literature Review

- Recording, summarizing and evaluating the existing findings and being able to provide expert judgment of the methods and finding is also called as literature review.
- Literature survey helps you to find out many answers while pursuing the research. It can be time consuming in initial stage.
- But if literature survey is not done in initial stage, then you might get stuck up while pursuing the research.
- Some authors have also given the meaning of literature review as the critical study of literature survey.

Literature Review

- A researcher cannot progress in the area of interest without the literature survey.

A literature survey can be divided into three parts

- Searching of the literature in the area of interest
- Collection/acquisition of literature or data
- Critical study of literature/review of literature in the area of interest
- Literature review helps you to widen the knowledge in the selected domain.
- It is very important to know the contributions of other researchers in the existing area or domain.

Literature Review

- Moreover, questions like, what challenges, which problems, what advantageous and disadvantageous of the existing systems, should trigger the thought process.
- These questions help to focus on finding answers to those limitations and accordingly you can define your own problem statement.
- Literature survey and literature review should go hand in hand. After searching the material available in the selected area of research, researcher has to read it, understand it, and note down the important points.
- Again, find the latest literature, read it, understand it, and note important points. This process of literature survey and literature review should be continuous.

Literature Review

- The literature survey is a comprehensive study of technical and authorized content related to research keywords.
- It is very important step in the initial phase of research, however, this step is revisited by researcher number of times during research journey.
- Literature survey provide details of research progress of particular domain.
- It helps the researcher to understand the approaches, methodologies, algorithms, and datasets used by other scientists.

Literature Review

- Also, it is important to identify where the gap is. Where can I contribute to an existing knowledgebase?
- It helps the researcher to understand the progress of domain and state of the art in the domain.
- Literature survey also helps to avoid duplication of work.
- Researcher should always avoid reinventing the wheel; this is important if one of the purposes of the research is to get a PhD degree

Literature Review

` Isaac Newton said,

“If I have seen further than others, it is by standing upon the shoulders of giants.”

- Researchers should be able to use previously done work as foundation and propose new idea above that may be supporting or contradicting to previous work.

Literature Review

- Open Access journals are one of the good sources of research articles, one can also access articles from library subscription, some university libraries etc.
- Good literature sources are periodicals, conference proceedings, research reports, standards, thesis, dissertation reports, research reports, white papers, patents, reviews, textbooks, handbooks, encyclopaedia etc.
- There are different strategies to search literature. One needs to first identify the keywords related to the problem definition and searching articles using keywords is the more precise way to search article

Literature Review: Backward Reference Searching

- **Backward reference searching**, also known as *chain searching*, involves identifying and examining the references or works cited in an article. It is done for the following reasons:
 - Learn about the development of knowledge on a topic.
 - Study the origins and development of a theory, construct, or model of interest.
 - Identify experts, institutions or organizations that specialize in a topic of research.
- **Backward Author Searching.** The works of an author can be examined by doing a backward reference search. This allows a researcher to review an author's previous work and publications, and also the development of their research.

Literature Review: Backward Reference Searching

Tools for Backward Reference Searching

- **Google Scholar** – Free and easy; check references and related citations
- **Scopus** – Advanced database with detailed reference lists (paid)
- **Web of Science** – High-quality citation tracking for research analysis
- **JSTOR** – Access full papers and manually review references
- **PubMed** – Best for medical/life sciences references
- **Connected Papers** – Visual graphs to find foundational papers

Literature Review: Backward Reference Searching

Backward Reference Searching Tools (IT)

- Google Scholar
- IEEE Xplore
- ACM Digital Library
- Scopus
- Web of Science
- CiteSeerX
- DBLP
- JSTOR
- PubMed (*useful for IT in healthcare*)
- Connected Papers

Literature Review: Forward Reference Searching

- **Forward reference searching** is when a researcher identifies articles that cite an original article or work after it had been published. This type of search focuses on the publications created **after** an article's publication.
- Forward reference searching helps a researcher do the following:
 - Expand their knowledge on a topic by locating follow-up studies.
 - Identify new findings and developments on a topic
 - Examine and consider other methodologies or analyses used to study a topic.
- **Forward Author Searching.** Forward reference searching can also be done to find a particular author and his or her works. This allows a researcher to review an author's later or more current works, and also examine new developments

Literature Review: Forward Reference Searching

Forward Reference Searching Tools

- **Google Scholar** : Use the “**Cited by**” feature to see newer papers
- **Scopus**: Shows **documents that cite the selected paper**
- **Web of Science** : Offers **Times Cited** feature for forward tracking
- **IEEE Xplore** : Displays **papers that cite a specific article**
- **ACM Digital Library** : Provides citation links to newer research
- **Semantic Scholar** : Shows **influential citations and newer works**
- **Connected Papers** : Visual graph of **related and newer papers**
- **ResearchGate** : Sometimes shows **who cited a paper**

Literature Review:

Novice researchers can follow below steps for literature survey:

- First download 100–120 research papers from published papers of IEEE/ACM/ Wiley/Elsevier/Springer/T and F or some other good SCI Indexed Journals related to research topic and start reading it. Normally, we do this type of search with the help of keywords of research problems. Some of the search engines for the literature survey are Google Scholar, CiteSeer, and so on.

Literature Review:

Novice researchers can follow below steps for literature survey:

- Researchers can initially read only title, abstract and conclusion of each research paper and decide whether it is related to research topic. If yes, include in list, else skip it. From this step you will get to know “What has been done in this area and what can be done in future?” From this understanding one can come up with his/ her research problem statement.

Literature Review:

Novice researchers can follow below steps for literature survey:

- Read shortlisted papers in detail and start taking notes of different parameters considered, methodologies used, conclusions derived, and related mathematics/ theories used. Researchers can finalize research hypothesis after this step and also can write good survey paper.

Literature Review:

- Another approach for literature survey is selecting any one best paper related to particular research problem of your interest. Try to find/download those papers that are cited and included in the references section. This approach leads to a better literature survey of a particular research problem.
- During the literature survey process, researchers should be able to identify the top 10 researchers in his/her domain. He/she should find good universities where people are working on the same or similar problems. Literature survey is a step that is often visited during the research journey. Researchers should always look at what other top researchers are doing in respective area of research.

Literature Review:

- Researchers should provide main focus on comparing the approaches, Identifying the weaknesses and strengths in recent research articles in the subject. Each paper's study can be summarized in systematic manner.
- Researcher may add some more parameters to be studied about the research paper and can prepare a format for same to keep record of each paper.
- Summarization and representation of literature surveys is an important part in the theses, presentations and in journal writing. Literature survey can be represented in textual, tabular or graphical format using different parameters.

Literature Review:

Literature surveys can be done with respect to following points:

- What
- How (process schematic)/standard procedure
- Discussion on major steps involved
- Design criteria and performance measures
- Techniques currently in use
- Comparative analysis (table or any suitable tool to discover and list pros and cons/ strength/weakness/future scope of existing techniques)
- Scope for research/gap in research

Nepal Journals Online (NepJOL)

- **NepJOL** is an online platform that provides access to **research journals published in Nepal**.
- NepJOL is a **database of peer-reviewed academic journals** that allows users to:
 - Search research papers
 - Read abstracts and full-text articles
 - Explore journals across multiple disciplines
- It is managed by the **Tribhuvan University Central Library** and hosted internationally.

Nepal Journals Online (NepJOL)

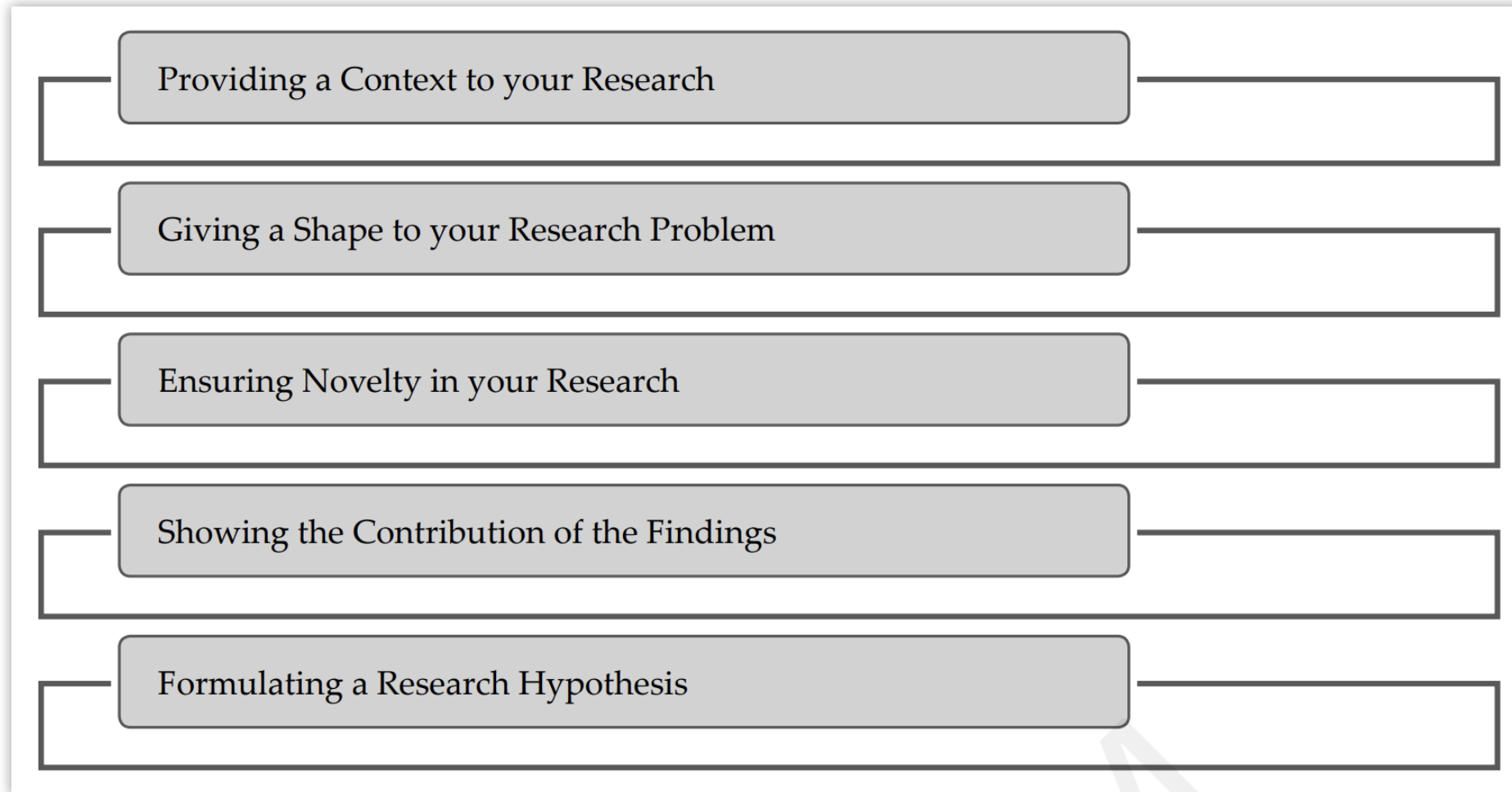
How NepJOL Contributes to Literature Survey

- **NepJOL** plays an important role in conducting a **literature survey (literature review)** by providing easy access to Nepalese research publications.
 - Provides Access to Local Research
 - Supports Finding Previous Studies
 - Helps Identify Research Gap
 - Improves Quality of Literature Review
 - Covers Multiple Disciplines

Function of Literature Review

- Literature review helps the researcher create a link or bond with the readers and build trust.
- It also helps eliminate the chances of the repetition of the similar research publication.
- It saves the time, money and other resources invested by the researcher in conducting research.
- A literature review gives a theoretical background of the research subject and establishes the relation between what a researcher is proposing to examine and what he/she has already studied.

Function of Literature Review



Function of Literature Review

- **Providing a context to your research:** A literature review helps place your research in the context of what is already known about the topic. It answers questions such as:
 - How does your research answer provide the solution to the management dilemma/question in comparison to the answers given by other researchers?
 - What contribution has your research work made?
 - What are the differences between your findings and the findings of other researchers?
- **Giving shape to your research problem:** By understanding the topic better, you will be able to conceptualise your research problem clearly and precisely. You will also be able to understand the relationship between your research problem and the work done in your research area.

Function of Literature Review

- **Ensuring novelty in your research:** Finally, a literature review ensures that you do not 'reinvent the wheel'. In other words, you save effort in trying to rediscover something that is already known or published. This will ensure that you bring new and significant contributions to your field of research.
- **Showing the contribution of the findings:** It enables you to show how your findings have contributed to the existing body of knowledge in your profession.
- **Formulating a research hypothesis:** Researchers read and review the available literature related to the research topic. The review may include reading articles, books, cases or other research papers. After the literature review is completed, the researchers gain a sufficient amount of information regarding their study topic which helps them in narrowing down or limiting it and expressing it in the form of a research question. The research hypothesis is constructed using the research question. Therefore, a literature review helps in formulating the research hypothesis.

Process of Literature Review

Search the existing literature
in your field of interest



Review the literature
obtained



Develop a theoretical
framework



Write up the literature
review

Process of Literature Review

Search the existing literature in your field of interest:

- First, search what has already been done in the chosen topic of interest. To search the existing literature, compile a bibliography and/or a list of references, which is a list of books on the topic of interest. To save time, the researcher can go through the following sources:
- Indices of journals on your research topic
- Abstracts of articles on your research topic
- Citation indices
- Digital libraries

Process of Literature Review

Review the literature obtained:

- After the researcher has identified the relevant journals and books, he/she must start reading them.
- Evaluate them critically to compile themes and issues that are associated with the research topic.
- The researcher must note down the main points to create a rough framework or theme of the research.

Process of Literature Review

Review the literature obtained:

Do a critical evaluation of the literature to:

- Identify the proposed theories, critics and methodologies (sample size, data used, measurement methods)
- Assess whether the knowledge relevant to your theoretical framework has been confirmed beyond doubt
- Discover different perspectives among researchers and write down your opinions about their validity
- Find the gaps that are present in the existing body of knowledge

Process of Literature Review

Develop a theoretical framework:

- A literature review can be a time-consuming task.
- Therefore, it is important to set the boundary and parameters for a research work.
- Sort out the information obtained from the literature sources according to theoretical framework.
- This will enable the researcher to focus in the literature search.
- In other words, the theoretical framework will provide foundation and guide to read further.
- It is quite possible as a researcher reads further. However, this is part of a research process.

Process of Literature Review

Write up the literature review:

- The final step is to compile and write all the literature read and reviewed. To do so, the researcher must perform the following steps:
 - a. Start review with a theme or points
 - b. Organize and list all the themes to discuss and relate. This will give a structure to literature review
 - c. Identify and describe various theories relevant to the field of research
 - d. Describe the gaps that exist in the body of knowledge in the field
 - e. Explain the recent advances and current trends in the field of research

Process of Literature Review

- f. Compare and evaluate findings based on:
 - i. Assumptions of research
 - ii. Theories related to the topic of research
 - iii. Hypotheses
 - iv. Research designs applied
 - v. Variables selected
 - vi. Potential future work speculated by the researchers
- g. Acknowledge, cite and quote sources of research. Give credit to the works of other researchers. Quote their work to show how research contradicts or contributes to their work. This will make the literature review more comprehensive and precise.

Research Question and Hypothesis

- The core part of your research starts with your research question or hypothesis and proceeds through your explanation of methods and results, or data analysis, and ends with a discussion of your findings.
- The research question and hypothesis mark your own disciplinary territory and drive your research.
- Whether you have both a research question and a hypothesis, and whether they are broken down into several sub-areas, will depend on your discipline and topic; but however they appear, it is important that they are clearly formulated.

Research Question

- Research questions are the foundation of any scholarly inquiry, guiding the direction and focus of the study.
- In IT research—where topics may include software development, cybersecurity, artificial intelligence, data analytics, and user experience—well-defined research questions are essential for identifying problems, guiding system design, and evaluating technological solutions.
- A strong research question in IT helps clarify the objectives of the study, whether it is to improve system performance, enhance security mechanisms, or analyze user interaction with digital platforms.

Research Question

- This section explores the concept and key characteristics of effective research questions within the IT domain, differentiates between exploratory questions (used to investigate emerging technologies or unknown problems) and descriptive questions (used to analyze existing systems or behaviors), and outlines strategies for developing precise, relevant, and researchable questions tailored to computing and technological contexts.
- **In conclusion :** A research question is a clearly formulated question that outlines the issue or problem your study aims to address. It sets the stage for the research design, data collection, and analysis, directing the inquiry toward a specific goal.

Research Question

- **Characteristics of a Good Research Question:**
 - **Clarity:** It should be clearly stated, avoiding ambiguity and ensuring that the research focus is understandable to others.
 - **Relevance:** The question should be significant to the field of study, addressing gaps in the literature or emerging issues in mass communications.
 - **Research ability:** It must be possible to answer the question through empirical investigation, using available research methods and tools.
 - **Specificity:** A good question is specific, targeting a particular aspect of the broader topic to make the research manageable and focused.

Research Question

Distinction Between Exploratory and Descriptive Research Questions

- **Exploratory Research Questions:** These questions are used when little is known about the topic or phenomenon. Exploratory questions aim to investigate and gain insights into a subject, seeking to understand how or why something happens. In exploratory question might ask, “**How do emerging social media platforms influence political engagement among young adults?**”
- **Descriptive Research Questions:** Descriptive questions aim to describe the characteristics or features of a subject. They are used when the goal is to provide an accurate representation or count of a phenomenon. A descriptive research question might be, “**What are the predominant themes in news coverage of environmental issues?**”

Research Question

Developing Clear and Focused Research Questions

- **Importance of Specificity and Feasibility:**
 - **Specificity:** Your research question should be narrowly tailored to address a specific issue within the broader field of Information Technology. This specificity helps in defining the study's scope and focusing the research efforts.
 - **Feasibility:** Consider the practical aspects of answering your research question, including the availability of data, time constraints, and resource limitations. A feasible question is one that can be realistically investigated within the parameters of your study.

Research Question

- **Strategies for Formulation:**

- **Literature Review:** Conduct a thorough review of existing research to identify gaps or unresolved questions in the field. This can inspire focused and relevant research questions.
- **Consultation:** Discuss your ideas with peers, mentors, or experts in mass communications. Feedback can help refine your questions and ensure they are both specific and feasible.
- **Pilot Studies:** Small-scale pilot studies or preliminary investigations can provide insights that help in formulating or refining your research questions.

Research Question

- Crafting clear and focused research questions is a critical step in the research process, setting the stage for meaningful and impactful inquiry.
- By ensuring that your questions are specific, feasible, and relevant to the field of mass communications, you lay the groundwork for a study that can contribute valuable insights to our understanding of Information Technology phenomena.

Research Question: Types

- In the pursuit of scientific inquiry within Information Technology, research questions serve as the navigational compass guiding the research process. These questions can be broadly categorized into two types: **nondirectional** and **directional**.
- Each type serves a distinct purpose and is formulated based on the nature of the study and the specific objectives the researcher aims to achieve. This section explores the definitions, uses, and strategies for crafting both nondirectional and directional research questions.

Research Question: Types

Nondirectional Research Questions

- **Definition:** Nondirectional research questions are open-ended queries that explore the existence of a relationship between variables without specifying the anticipated direction of this relationship. They are used when the literature does not strongly suggest which outcome is expected or when exploring new or under-researched areas.
- **When to Use Them:** Employ nondirectional questions when previous research is inconclusive, conflicting, or absent. They are particularly useful in exploratory studies where the aim is to uncover patterns, relationships, or phenomena without presupposing outcomes.

Research Question: Types

- **Crafting Questions:**

- **Focus on Exploration:** Phrase your question to emphasize exploration, such as “Is there a relationship between social media usage and political participation among young adults?”
- **Avoid Implied Direction:** Ensure the wording does not inadvertently suggest a presumed direction of the relationship. The question should remain open to any outcome, whether positive, negative, or neutral.

Research Question: Types

Directional Research Questions

- **Definition:** Directional research questions specify the expected direction of the relationship between variables. These questions are based on predictions that are often derived from theoretical frameworks or existing literature.
- **Purposes:** Directional questions are used when there is sufficient theoretical or empirical basis to hypothesize a particular outcome. They guide the research towards testing specific hypotheses, making them suitable for studies aiming to confirm or refute theoretical predictions.

Research Question: Types

- **Formulating Questions:**

- **Specify Expected Outcomes:** Clearly articulate the anticipated direction of the relationship in the question. For example, “Does increased exposure to environmental news lead to higher levels of environmental activism among viewers?”
- **Ground in Literature:** Ensure that the directionality implied by your question is supported by theoretical rationales or empirical evidence from previous research. This alignment strengthens the justification for expecting a particular outcome.

Research Question: Types

Strategies for Formulating Research Questions

- **Engage with Current Research:** Immerse yourself in the latest studies and debates within the field of study to identify trends, gaps, and areas ripe for investigation.
- **Consult Theoretical Frameworks:** Draw on established theories to guide the formulation of your questions, whether seeking to explore uncharted territory (nondirectional) or test specific propositions (directional).
- **Iterative Refinement:** Research questions often evolve during the initial stages of a study. Be prepared to refine your questions as you delve deeper into the literature and sharpen your study's focus.

Research Question

Example research questions:

- What impact will climate change have on production systems in the Tasmanian dairy industry?
- What are the unique challenges and opportunities for higher education institutions that allow first-year undergraduate students to use Generative AI tools in written assessments?

Research Hypothesis

- A hypothesis is a well-reasoned proposition in response to a research question that you will test to confirm or disprove in your research.
- Not all research has a hypothesis.
A hypothesis should:
 - Be in the form of a statement
 - Be disprovable
 - Be clear and specific in scope.

Research Hypothesis

- **Definition:** A hypothesis is a predictive statement that proposes a possible outcome or relationship between two or more variables.
- It is grounded in theory or prior empirical findings and serves as a basis for scientific inquiry.
- **Function:**
 - The primary function of a hypothesis is to provide a specific, testable proposition derived from the broader research question.
 - Hypotheses guide the **research design, data collection, and analysis process**, offering a clear focus for **empirical investigation**.
 - They enable researchers to apply statistical methods to test the proposed relationships or effects, thereby contributing to the accumulation of scientific knowledge.

Research Hypothesis

The Relationship Between Research Questions and Hypotheses

- **From Questions to Hypotheses:** Research questions set the stage for the research by identifying the **key phenomena** or **relationships of interest**. Hypotheses take this a step further by specifying the expected direction or nature of these relationships based on theoretical or empirical groundwork. Essentially, while research questions identify “what” the study aims to explore, hypotheses propose “how” these explorations will unfold.
- **Complementarity:** Research questions and hypotheses are complementary, with the former providing a broad inquiry framework and the latter offering a focused, conjectural answer that can be empirically tested. This synergy ensures that the research is both guided by curiosity and anchored in a framework that facilitates systematic investigation.

Research Hypothesis: Types of Hypotheses

Null Hypotheses (H0)

- **Definition:** The null hypothesis (H0) posits that there is no difference, effect, or relationship between the variables under investigation. It represents a statement of skepticism or neutrality, suggesting that any observed differences or relationships in the data are due to chance rather than a systematic effect.
- **Role in Research:** The null hypothesis serves as a benchmark for testing the existence of an effect or relationship. By attempting to disprove or reject the null hypothesis through statistical analysis, researchers can provide evidence supporting the presence of a meaningful effect or relationship. The null hypothesis is foundational in hypothesis testing, enabling researchers to apply statistical methods to determine the likelihood that observed data could have occurred under the null condition.
- **Formulating Null Hypotheses:** Null hypotheses are formulated as statements of no difference or no relationship. For example, in a study examining the impact of social media usage on political engagement, a null hypothesis might state, “There is no difference in political engagement levels between users and non-users of social media.”

Research Hypothesis: Types of Hypotheses

Alternative Hypotheses (H1)

- **Definition:** The alternative hypothesis (H1) is the counter proposition to the null hypothesis. It posits that there is a significant difference, effect, or relationship between the variables being studied. The alternative hypothesis reflects the researcher's theoretical expectation or prediction about the outcome of the study.
- **Complementing Null Hypotheses:** The alternative hypothesis directly complements the null hypothesis by specifying the expected effect or relationship that the research aims to demonstrate. While the null hypothesis posits the absence of an effect, the alternative hypothesis asserts its presence, guiding the direction of the study's empirical investigation.
- **Crafting Alternative Hypotheses:** Alternative hypotheses are crafted to predict specific outcomes based on the research question and theoretical framework. They should clearly articulate the anticipated direction or nature of the relationship or difference between variables. Continuing the earlier example, an alternative hypothesis might state, "Users of social media exhibit higher levels of political engagement than non-users."

Research Hypothesis: Types of Hypotheses

Strategic Formulation of Hypotheses

- The formulation of null and alternative hypotheses is a strategic exercise that sets the stage for empirical testing. Effective hypotheses are:
- **Specific and Concise:** Clearly define the variables and the expected relationship or difference, avoiding ambiguity.
- **Empirically Testable:** Ensure that the hypotheses can be tested using available research methods and data.
- **Theoretically Grounded:** Base your hypotheses on existing literature, theories, or preliminary evidence, providing a rationale for the expected outcomes.

Example:

Impact of Class Size on Academic Performance

Research Question: How does class size affect students' academic performance in primary schools?

Hypothesis: Smaller class sizes will result in higher academic performance compared to larger classes.

SMART Objectives:

- Specific: Measure the impact of class size on academic performance.
- Measurable: Compare average test scores between small and large classes.
- Achievable: Conduct the study within a semester.
- Relevant: Inform educational policy on class size.
- Time-bound: Complete data collection within three months.

Example:

Research Topic

Impact of Social Media Usage on Academic Performance of Undergraduate Students

Research Problem

Despite the widespread use of social media platforms among students, its effect on academic performance remains unclear. Some students benefit from collaborative learning, while others experience distraction and reduced productivity. There is a need to systematically analyze whether social media usage positively or negatively impacts academic outcomes.

Example:

Research Objectives

- To analyze the pattern of social media usage among undergraduate students
- To evaluate the relationship between social media usage and academic performance
- To identify whether social media acts as a learning tool or a distraction
- To suggest strategies for effective use of social media in education

Research Questions

- How frequently do students use social media for academic purposes?
- What is the relationship between time spent on social media and academic performance?
- Does social media usage improve collaborative learning among students?
- What are the negative impacts of excessive social media usage on studies?

Example:

Hypothesis

- **Null Hypothesis (H_0):**

There is **no significant relationship** between social media usage and academic performance of students.

- **Alternative Hypothesis (H_1):**

There **is a significant relationship** between social media usage and academic performance of students.

Example:

Research Topic

- **“Performance Analysis of Symmetric Key Encryption Algorithms for Secure Data Transmission”**

Research Problem

- With the increasing need for secure data transmission in modern applications, symmetric key encryption algorithms such as AES (Advanced Encryption Standard), DES (Data Encryption Standard), and Blowfish are widely used. However, there is a challenge in selecting the most efficient algorithm in terms of security, speed, and resource utilization. A systematic comparison is required to determine which algorithm performs best under different conditions.

Example:

Research Objectives

- To study different symmetric key encryption algorithms
- To compare the performance of AES, DES, and Blowfish
- To evaluate encryption and decryption time for each algorithm
- To analyze security strength and resource consumption
- To recommend the most suitable algorithm for secure data transmission

- **Research Questions**

- Which symmetric key encryption algorithm provides the best performance?
- How do AES, DES, and Blowfish differ in terms of speed and security?
- What is the impact of key size on encryption efficiency?
- Which algorithm is most suitable for real-time applications?

Example:

Hypothesis

- **Null Hypothesis (H_0):**

There is **no significant difference** in performance among symmetric key encryption algorithms.

- **Alternative Hypothesis (H_1):**

There **is a significant difference** in performance among symmetric key encryption algorithms.

Example:

Research Topic

- **“Comparative Analysis of AES-128, AES-192, and AES-256 Based on Performance and Security”**

Research Problem

- Although AES (Advanced Encryption Standard) is widely adopted for secure data encryption, its different key-size variants (AES-128, AES-192, and AES-256) offer varying levels of security and performance. Selecting the most appropriate variant for specific applications (e.g., real-time systems vs high-security systems) remains a challenge. A systematic comparison is required to evaluate trade-offs between speed, resource usage, and security strength.

Example:

Research Objectives

- To study the working principles of AES-128, AES-192, and AES-256
- To compare encryption and decryption performance of each variant
- To analyze memory and CPU usage during execution
- To evaluate security strength based on key size
- To recommend the most suitable AES variant for different use cases

Research Questions

- What are the performance differences between AES-128, AES-192, and AES-256?
- How does key size affect encryption and decryption time?
- Which AES variant provides the best balance between speed and security?
- Which variant is most suitable for resource-constrained environments?

Example:

Hypothesis

- **Null Hypothesis (H_0):**

There is **no significant difference** in performance and security among AES-128, AES-192, and AES-256.

- **Alternative Hypothesis (H_1):**

There **is a significant difference** in performance and security among AES-128, AES-192, and AES-256.

Example:

Research Topic

- **“Security Factor Analysis of AES-128, AES-192, and AES-256 Encryption Algorithms”**

Research Problem

- The AES (Advanced Encryption Standard) algorithm is widely used for securing sensitive data, but its variants (AES-128, AES-192, and AES-256) provide different levels of security due to variations in key size and number of encryption rounds. Organizations often face difficulty in selecting the appropriate variant that ensures sufficient protection against modern cryptographic attacks while maintaining efficiency. A detailed analysis of security factors is required to identify the most robust variant.

Example:

Research Objectives

- To analyze the security mechanisms of AES-128, AES-192, and AES-256
- To evaluate resistance against common cryptographic attacks (brute force, differential, linear cryptanalysis)
- To compare key strength and number of rounds among AES variants
- To assess vulnerability related to implementation issues (e.g., side-channel attacks)
- To recommend the most secure AES variant for different security requirements

Example:

Research Questions

- How does key size influence the security strength of AES variants?
- Which AES variant is more resistant to brute-force and cryptanalytic attacks?
- What are the major vulnerabilities associated with AES implementations?
- Is AES-256 significantly more secure than AES-128 in practical scenarios?

Hypothesis

- **Null Hypothesis (H_0):**

There is **no significant difference** in the security strength of AES-128, AES-192, and AES-256.

- **Alternative Hypothesis (H_1):**

There is a **significant difference** in the security strength of AES-128, AES-192, and AES-256.

Example: